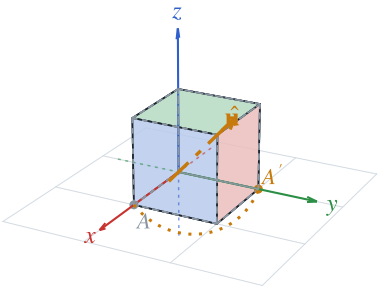
IB MATHEMATICS · A PROJECT ON MATRICES

# Rotation in Space

Rotation matrices in three dimensions:
from the unit vectors to the centre of rotation

This report follows a project in which an IB student investigated rotations in space using matrices. The project begins with the three unit vectors and a cube with one corner at the origin. It ends with a matrix that rotates any shape about any axis through any point, and with the inverse problem: when the start and end positions are known, the axis, the angle and the centre of rotation can be found.
The report contains theory, derivations, worked examples and ten exercises with answers. It goes together with an interactive 3D page on which every example can be played back.




Michael H. Grant
Senior lecturer emeritus in physics & mathematics, Grenaa Gymnasium, Denmark
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**Conventions**

The coordinate system is right-handed with the $z$-axis pointing up. The unit vectors are $\mathbf{i} = (1,0,0)$, $\mathbf{j} = (0,1,0)$ and $\mathbf{k} = (0,0,1)$. Points and vectors are written as columns, and a matrix acts from the left: $P' = RP$. A positive angle turns anticlockwise when seen from the tip of the axis looking back towards the origin (right-hand rule: thumb along the axis, fingers curl in the direction of rotation).

## 1 The unit vectors determine the matrix

A rotation about an axis through the origin is a *linear* map: it preserves sums and multiples, $R(a\mathbf{v}+b\mathbf{w}) = aR\mathbf{v}+bR\mathbf{w}$. Every vector can be written $\mathbf{v} = x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$, so

$$R\mathbf{v} = x\,R\mathbf{i} + y\,R\mathbf{j} + z\,R\mathbf{k}.$$

The map is therefore completely determined by the images of the three unit vectors. Placing them side by side as columns gives the matrix:

**The column rule**

$$R = \begin{bmatrix} | & | & | \\ R\mathbf{i} & R\mathbf{j} & R\mathbf{k} \\ | & | & | \end{bmatrix}.$$

Column $n$ of a matrix is the image of the $n$th unit vector.

### 1.1 Rotation about the $z$-axis

When $\mathbf{i}$ is turned through the angle $\theta$ in the $xy$-plane, it lands at $(\cos\theta, \sin\theta, 0)$. The vector $\mathbf{j}$ is $90^\circ$ ahead and lands at $(\cos(\theta+90^\circ), \sin(\theta+90^\circ), 0) = (-\sin\theta, \cos\theta, 0)$. The vector $\mathbf{k}$ lies on the axis and does not move. The column rule gives

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

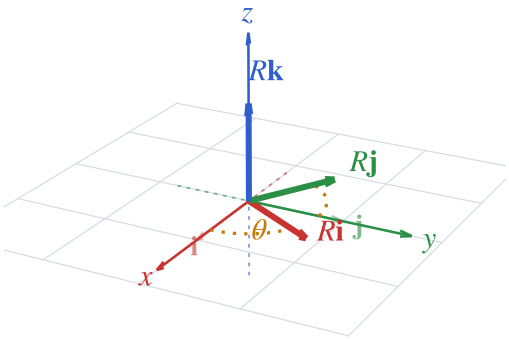



**Figure 1.** The unit vectors turned through $\theta = 60^\circ$ about the $z$-axis. The faint arrows are $\mathbf{i}$ and $\mathbf{j}$ before the rotation; the bold ones are their images. $R\mathbf{k} = \mathbf{k}$.

### 1.2 Rotation about the $x$- and $y$-axes

In the same way one finds

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \qquad R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}.$$

Notice that $-\sin\theta$ in $R_y$ sits in the *bottom* left. This comes from the cyclic order $x \to y \to z \to x$. About the $x$-axis the positive direction turns $y$ towards $z$, about the $z$-axis it turns $x$ towards $y$, and about the $y$-axis it turns $z$ towards $x$, not $x$ towards $z$. Hence $R_y(\theta)\mathbf{i} = (\cos\theta, 0, -\sin\theta)$: for small positive angles $\mathbf{i}$ dips below the $xy$-plane.

## 2 The cube with a corner at the origin

The cube has corners $(x, y, z)$ with $x, y, z \in \{0, 1\}$. Each corner is a position vector $P$, and its image is $P' = RP$. Since the axis passes through the origin, the corner $(0, 0, 0)$ stays put.

**Example.** A rotation of $90^\circ$ about the $x$-axis. With $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$,

$$R_x(90^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad R_x(90^\circ) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ -z \\ y \end{bmatrix}.$$

| $P$ | | $P'$ |
|---|---|---|
| $(0,0,0)$ | $\longmapsto$ | $(0,0,0)$ |
| $(1,0,0)$ | $\longmapsto$ | $(1,0,0)$ |
| $(1,1,0)$ | $\longmapsto$ | $(1,0,1)$ |
| $(0,1,0)$ | $\longmapsto$ | $(0,0,1)$ |
| $(0,0,1)$ | $\longmapsto$ | $(0,-1,0)$ |
| $(1,0,1)$ | $\longmapsto$ | $(1,-1,0)$ |
| $(1,1,1)$ | $\longmapsto$ | $(1,-1,1)$ |
| $(0,1,1)$ | $\longmapsto$ | $(0,-1,1)$ |

**Table 1.** The corners of the cube before and after $R_x(90^\circ)$.

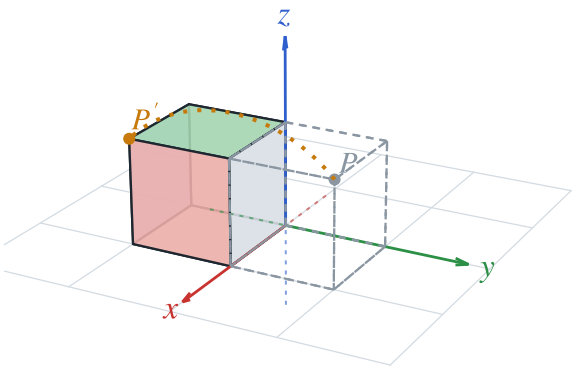



**Figure 2.** The cube turned $90^\circ$ about the $x$-axis. The corner $P = (1, 1, 1)$ follows a quarter circle in the plane $x = 1$.

The corners with $x = 1$ lie in the plane $x = 1$ both before and after. That is a useful check: a rotation about the $x$-axis never changes the $x$-coordinate.

## 3 Composite rotations

Rotating first by $R_1$ and then by $R_2$ gives the image $R_2(R_1P) = (R_2R_1)P$. The combined matrix is the product, and *the rotation performed first stands furthest to the right*.

**Order matters.** With $R_x = R_x(90^\circ)$ and $R_y = R_y(90^\circ)$,

$$R_xR_y = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad R_yR_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}.$$

The columns show that $R_x R_y \mathbf{i} = (0,1,0)$, whereas $R_y R_x \mathbf{i} = (0,0,-1)$. Matrix multiplication is not commutative, and you can see it with a matchbox in your hand.

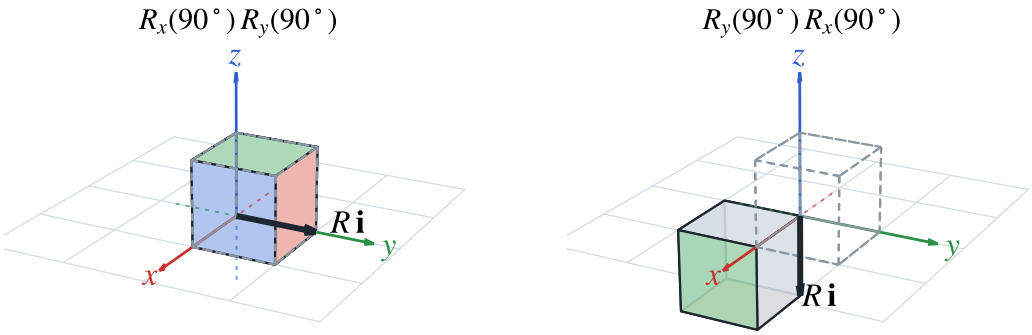



**Figure 3.** The same two quarter turns in each order. On the left, first about $y$, then about $x$. On the right, the other way round. The cube and the image of $\mathbf{i}$ end up in different places.

### 3.1 Properties of rotation matrices

The columns of a rotation matrix are the images of $\mathbf{i}, \mathbf{j}, \mathbf{k}$. A rotation preserves lengths and angles, so the columns have length 1 and are perpendicular to each other. In short:

**Orthogonality**

$$R^{\mathsf{T}} R = I, \qquad R^{-1} = R^{\mathsf{T}}, \qquad \det R = 1.$$

The inverse rotation is simply the transpose, which is far cheaper to compute than an ordinary inverse. The condition $\det R = 1$ separates rotations from reflections, which also satisfy $R^{\mathsf{T}} R = I$ but have $\det R = -1$. The rotation matrices form the group $SO(3)$.

## 4 Rotation about an arbitrary axis through the origin

We now rotate through the angle $\theta$ about an axis with direction vector $\mathbf{u}$. First normalise it: $\hat{\mathbf{u}} = \mathbf{u}/|\mathbf{u}|$.

### 4.1 Derivation by decomposition

Split a vector $\mathbf{v}$ into a part along the axis and a part perpendicular to it:

$$\mathbf{v}_{\parallel} = (\mathbf{v} \cdot \hat{\mathbf{u}})\,\hat{\mathbf{u}}, \qquad \mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}.$$

The part along the axis does not move. The perpendicular part turns in the plane perpendicular to $\hat{\mathbf{u}}$. In that plane, $\mathbf{v}_{\perp}$ and $\hat{\mathbf{u}} \times \mathbf{v}$ are perpendicular vectors of equal length, and $\hat{\mathbf{u}} \times \mathbf{v}$ is $90^\circ$ ahead in the direction of rotation. Therefore

$$R\mathbf{v} = \mathbf{v}_{\parallel} + \cos\theta\, \mathbf{v}_{\perp} + \sin\theta\, (\hat{\mathbf{u}} \times \mathbf{v}).$$

Writing each term as a matrix times $\mathbf{v}$, with $\mathbf{v}_{\parallel} = \hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}}\mathbf{v}$ and $\hat{\mathbf{u}} \times \mathbf{v} = [\hat{\mathbf{u}}]_{\times}\mathbf{v}$, gives Rodrigues' formula:

**Rodrigues' rotation formula**

$$R = \cos\theta\, I + \sin\theta\, [\hat{\mathbf{u}}]_\times + (1-\cos\theta)\, \hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}}, \qquad [\hat{\mathbf{u}}]_\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}.$$

Put $\hat{\mathbf{u}} = \mathbf{k}$ and check that the formula gives $R_z(\theta)$.

## 4.2 Derivation by change of basis

Another route, which many students find more natural: choose a right-handed orthonormal basis $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{u}})$ and let $B$ have these vectors as its columns. In that basis the rotation is just $R_z(\theta)$. Change to the new basis with $B^{\mathsf{T}}$, rotate, and change back with $B$:

$$R = B\, R_z(\theta)\, B^{\mathsf{T}}.$$

For the axis $(1,1,1)$ one can use, for example, $\hat{\mathbf{e}}_1 = \frac{1}{\sqrt{2}}(1,-1,0)$ and $\hat{\mathbf{e}}_2 = \hat{\mathbf{u}} \times \hat{\mathbf{e}}_1 = \frac{1}{\sqrt{6}}(1,1,-2)$. Both routes give the same matrix.

**Example.** $120^\circ$ about the space diagonal $\mathbf{u} = (1,1,1)$. Here $\hat{\mathbf{u}} = \frac{1}{\sqrt{3}}(1,1,1)$, $\cos\theta = -\frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$, so

$$(1-\cos\theta)\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad \sin\theta\,[\hat{\mathbf{u}}]_\times = \frac{1}{2}\begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}.$$

Adding $-\frac{1}{2}I$ gives

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

This is a cyclic permutation of the axes: $\mathbf{i} \to \mathbf{j} \to \mathbf{k} \to \mathbf{i}$. It is also exactly $R_x R_y$ from Section 3. Two quarter turns about two different axes give a third of a turn about a third axis.

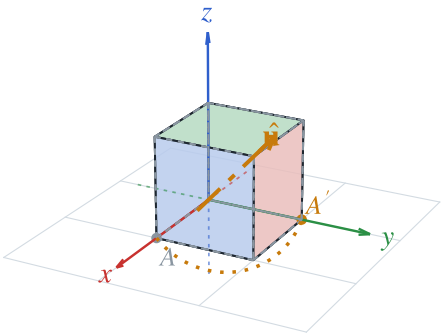



**Figure 4.** The cube turned $120^\circ$ about the space diagonal. The corner $A = (1,0,0)$ goes to $A' = (0,1,0)$, and the cube lands on itself.

**The trace.** The sum of the diagonal, $\operatorname{tr} R$, will be useful later. Since $\operatorname{tr} I = 3$, $\operatorname{tr}[\hat{\mathbf{u}}]_\times = 0$ and $\operatorname{tr}(\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}}) = |\hat{\mathbf{u}}|^2 = 1$,

$$\operatorname{tr} R = 1 + 2\cos\theta.$$

## 5 An arbitrary centre of rotation

If the axis passes through a point $C$ rather than the origin, first move $C$ to the origin, rotate, and move back:

$$P' = R(P - C) + C = RP + (C - RC).$$

This is a linear part $R$ plus a constant translation $\mathbf{t} = C - RC$. Such a map cannot be written as a $3 \times 3$ matrix, because it does not send the origin to the origin.

### 5.1 Homogeneous coordinates

The trick from computer graphics is to add a fourth coordinate and write the point $(x, y, z)$ as $(x, y, z, 1)$. Then a translation becomes a matrix too:

$$T(\mathbf{a}) = \begin{bmatrix} 1 & 0 & 0 & a_x \\ 0 & 1 & 0 & a_y \\ 0 & 0 & 1 & a_z \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \widetilde{R} = \begin{bmatrix} R & \mathbf{0} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}.$$

**Rotation about the axis through $C$**

$$M = T(C)\,\widetilde{R}\,T(-C) = \begin{bmatrix} R & C - RC \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}.$$

Read from the right: move $C$ to the origin, rotate, move back. The last column is the translation $\mathbf{t} = C - RC$.

**Example.** $90^\circ$ about an axis parallel to the $z$-axis through $C = (1, 1, 0)$. Here $R = R_z(90^\circ)$ and $RC = (-1, 1, 0)$, so $\mathbf{t} = (2, 0, 0)$ and

$$M = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The point $(2, 1, 0)$ goes to $(-1 + 2,\ 2,\ 0) = (1, 2, 0)$. Check: $(2, 1, 0) - C = (1, 0, 0)$ turns to $(0, 1, 0)$, and adding $C$ gives $(1, 2, 0)$.

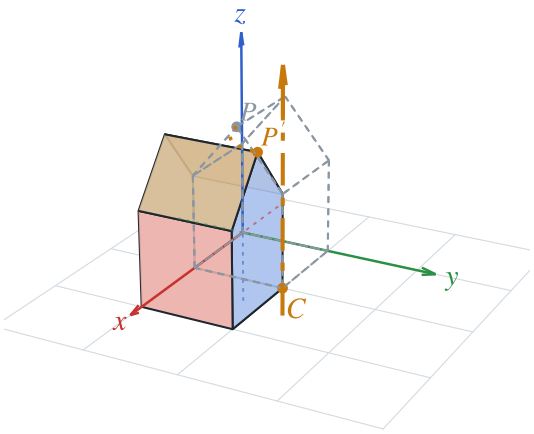



**Figure 5.** A house turned 90° about a vertical axis through $C = (1,1,0)$. A house is used because it is not symmetric, so you can see which way it faces.

## 6 The inverse problem

Now we know the start and end positions and must find the rotation. We follow four corners of the cube,

$$O = (0,0,0), \quad A = (1,0,0), \quad B = (0,1,0), \quad D = (0,0,1),$$

and know their final positions $O', A', B', D'$.

### 6.1 Step 1: the matrix

The map is $P' = RP + \mathbf{t}$. Subtracting two images removes the translation: $A' - O' = R(A - O) = R\mathbf{i}$. The column rule gives at once

$$R = \begin{bmatrix} | & | & | \\ A' - O' & B' - O' & D' - O' \\ | & | & | \end{bmatrix}, \qquad \mathbf{t} = O'.$$

If instead three arbitrary non-coplanar edge vectors are known, collected as the columns of $K$ before and $K'$ after, then $R = K'K^{-1}$.

**Check.** Verify that $R^{\mathsf{T}}R = I$ and $\det R = 1$. Otherwise this is not a rotation, perhaps because the positions were measured wrongly or because the shape has been reflected.

### 6.2 Step 2: the angle

From the trace: $\cos\theta = \dfrac{\operatorname{tr} R - 1}{2}$, with $0 \le \theta \le 180°$.

### 6.3 Step 3: the axis

In Rodrigues' formula $I$ and $\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}}$ are symmetric, while $[\hat{\mathbf{u}}]_\times$ is antisymmetric. Hence $R - R^{\mathsf{T}} = 2\sin\theta\,[\hat{\mathbf{u}}]_\times$, and the axis can be read off:

$$\hat{\mathbf{u}} = \frac{1}{2\sin\theta}\begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix}.$$

If $\theta = 180^\circ$ then $\sin\theta = 0$. In that case $R$ is symmetric, and one uses $\frac{1}{2}(R + I) = \hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}}$ instead: any column with a non-zero diagonal element is parallel to $\hat{\mathbf{u}}$.

### 6.4 Step 4: the centre of rotation

From Section 5 we know that $\mathbf{t} = C - RC$, that is,

$$(I - R)\,C = \mathbf{t}.$$

The matrix $I - R$ is singular, because every point on the axis is fixed. This agrees with the geometry: if $C$ is a solution, so is $C + s\hat{\mathbf{u}}$, and the solution set is the whole axis. We choose the point $C_0$ on the axis closest to the origin, that is, with $C_0 \cdot \hat{\mathbf{u}} = 0$.

For a vector $\mathbf{w} \perp \hat{\mathbf{u}}$, Rodrigues' formula says $R\mathbf{w} = \cos\theta\,\mathbf{w} + \sin\theta\,\hat{\mathbf{u}} \times \mathbf{w}$. The equation becomes

$$(1 - \cos\theta)\,C_0 - \sin\theta\,\hat{\mathbf{u}} \times C_0 = \mathbf{t}.$$

The operation $J\mathbf{w} = \hat{\mathbf{u}} \times \mathbf{w}$ turns through $90^\circ$ in the plane and satisfies $J^2 = -1$, just like $i$ in the complex numbers. With $a = 1 - \cos\theta$ and $b = \sin\theta$ the equation reads $(a - bJ)C_0 = \mathbf{t}$. Multiplying by $a + bJ$ and using $a^2 + b^2 = 2(1 - \cos\theta)$ gives

**The centre closest to the origin**

$$C_0 = \frac{(1 - \cos\theta)\,\mathbf{t} + \sin\theta\,(\hat{\mathbf{u}} \times \mathbf{t})}{2(1 - \cos\theta)}.$$

### 6.5 When there is no centre of rotation

The formula assumes $\mathbf{t} \cdot \hat{\mathbf{u}} = 0$. If $\mathbf{t}$ has a component along the axis, the shape has also been pushed along the axis, and $(I - R)C = \mathbf{t}$ has no solution. By *Chasles' theorem*, every orientation-preserving rigid motion in space is a *screw motion*: a rotation about an axis combined with a translation along the same axis. A pure rotation is the case where that translation is zero.

**Worked example.** After a rotation the corners lie at $O' = (2,0,0)$, $A' = (2,1,0)$, $B' = (1,0,0)$ and $D' = (2,0,1)$.

1. The edge vectors are $(0,1,0)$, $(-1,0,0)$ and $(0,0,1)$, so $R = R_z(90^\circ)$, and $\mathbf{t} = (2,0,0)$.
2. $\operatorname{tr} R = 1$ gives $\cos\theta = 0$, so $\theta = 90^\circ$.
3. $R - R^{\mathsf{T}}$ gives $(0,0,2)/(2 \cdot 1)$, so $\hat{\mathbf{u}} = (0,0,1)$.
4. $\mathbf{t} \cdot \hat{\mathbf{u}} = 0$ and $\hat{\mathbf{u}} \times \mathbf{t} = (0,2,0)$. The formula gives $C_0 = \big((2,0,0) + (0,2,0)\big)/2 = (1,1,0)$.

This is exactly the situation from Section 5, now seen in reverse.

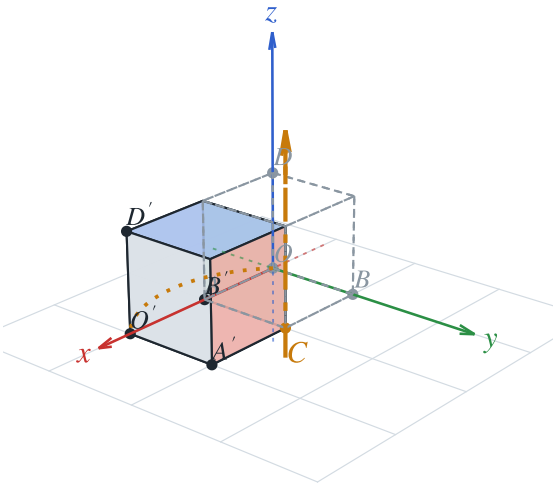



**Figure 6.** The inverse problem. The dashed cube is the start, the coloured one the end. The axis found runs vertically through $C = (1, 1, 0)$, and the corner $O$ follows the dotted quarter circle to $O'$.

## 7 Further directions

Rotation matrices are used wherever things turn in space: in computer games and 3D graphics, in robot arms, in satellite attitude control and in crystallography. A natural next step for a student is *Euler angles*, where an arbitrary rotation is written as three rotations about coordinate axes, and the problem of *gimbal lock*, where two of the three axes coincide. Another is *quaternions*, which describe a rotation with four numbers instead of nine and are standard in modern graphics.

## 8 Exercises

**Exercise 1 — A quarter turn about the $z$-axis**

Write down $R_z(90^\circ)$ and find the images of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

**Exercise 2 — Where does the corner go?**

The cube with a corner at the origin is turned $90^\circ$ about the $x$-axis. Find the images of the corners $(1, 1, 1)$ and $(0, 1, 0)$.

**Exercise 3 — Order matters**

Compute $R_x(90^\circ)R_y(90^\circ)$ and $R_y(90^\circ)R_x(90^\circ)$. Where does $\mathbf{i}$ end up in each case?

**Exercise 4 — Rotation about the space diagonal**

Use Rodrigues' formula to find the matrix of a rotation of $120^\circ$ about the axis $\mathbf{u} = (1, 1, 1)$.

**Exercise 5 An axis through another point**

Rotate $90^\circ$ about an axis parallel to the $z$-axis through $C = (1,1,0)$. Find the images of $(2,1,0)$ and $(1,0,3)$, and write down the $4 \times 4$ matrix $M$.

**Exercise 6 Find the centre of rotation**

After a rotation the corners of the cube lie at $O' = (2,0,0)$, $A' = (2,1,0)$, $B' = (1,0,0)$ and $D' = (2,0,1)$. Find $R$, the angle, the axis and the centre of rotation.

**Exercise 7 Read off the axis and the angle**

A rotation about an axis through the origin has the matrix $R = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Find the angle and the axis.

**Exercise 8 From a $4 \times 4$ matrix to the centre of rotation**

The map $M = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ has the rotation part from Exercise 7 and the translation $\mathbf{t} = (1,-1,0)$. Find a point $C$ on the axis of rotation.

**Exercise 9 Is it a rotation at all?**

The final positions are $O' = (0,0,1)$, $A' = (0,1,1)$, $B' = (-1,0,1)$ and $D' = (0,0,2)$. Is there a centre of rotation?

**Exercise 10 Prove two properties**

Show that $R_z(\alpha)R_z(\beta) = R_z(\alpha+\beta)$, and that $R^\mathsf{T}R = I$ and $\det R = 1$ for $R = R_z(\theta)$.

## 9 Answers

**Exercise 1.** $R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The columns can be read off directly: $\mathbf{i} \mapsto \mathbf{j}$, $\mathbf{j} \mapsto -\mathbf{i}$ and $\mathbf{k} \mapsto \mathbf{k}$.

**Exercise 2.** $R_x(90^\circ)(x, y, z) = (x, -z, y)$. So $(1, 1, 1) \mapsto (1, -1, 1)$ and $(0, 1, 0) \mapsto (0, 0, 1)$.

**Exercise 3.**

$$R_x R_y = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad R_y R_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}.$$

$R_x R_y \mathbf{i} = (0, 1, 0)$, but $R_y R_x \mathbf{i} = (0, 0, -1)$. The products are different.

**Exercise 4.** $\hat{\mathbf{u}} = \frac{1}{\sqrt{3}}(1, 1, 1)$, $\cos\theta = -\frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$. The term $(1 - \cos\theta)\hat{\mathbf{u}}\hat{\mathbf{u}}^\mathsf{T}$ has $\frac{1}{2}$ in every entry, and $\sin\theta[\hat{\mathbf{u}}]_\times$ has $\pm\frac{1}{2}$ off the diagonal. Adding $-\frac{1}{2}I$ gives $R = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, the cyclic permutation $x \to y \to z \to x$. It is the same matrix as $R_x R_y$ in Exercise 3.

**Exercise 5.** $(2, 1, 0) - C = (1, 0, 0) \mapsto (0, 1, 0)$, and adding $C$ gives $(1, 2, 0)$. $(1, 0, 3) - C = (0, -1, 3) \mapsto (1, 0, 3)$, and adding $C$ gives $(2, 1, 3)$. With $C - RC = (1, 1, 0) - (-1, 1, 0) = (2, 0, 0)$,

$$M = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

**Exercise 6.** $R = R_z(90^\circ)$ and $\mathbf{t} = (2, 0, 0)$. $\operatorname{tr} R = 1$ gives $\theta = 90^\circ$, and the axis is $(0, 0, 1)$. The equation $(I - R)C = \mathbf{t}$ gives $x + y = 2$ and $-x + y = 0$, so $C = (1, 1, 0)$ plus any amount along the $z$-direction.

**Exercise 7.** $\operatorname{tr} R = 0$, so $\cos\theta = -\frac{1}{2}$ and $\theta = 120^\circ$. The antisymmetric part gives $(R_{32} - R_{23},\ R_{13} - R_{31},\ R_{21} - R_{12}) = (1, 1, 1)$, so $\hat{\mathbf{u}} = \frac{1}{\sqrt{3}}(1, 1, 1)$.

**Exercise 8.** The axis is $\frac{1}{\sqrt{3}}(1, 1, 1)$ and $\theta = 120^\circ$. First check that $\mathbf{t} \cdot \hat{\mathbf{u}} = 0$, so this is a pure rotation. $(I - R)C = \mathbf{t}$ gives

$$x - z = 1, \qquad -x + y = -1, \qquad -y + z = 0.$$

The solution is the line $C = (1, 0, 0) + s(1, 1, 1)$. The point closest to the origin is $C_0 = \left(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}\right)$, which also follows from the formula in Section 6.4: $\frac{1}{3}\left(\frac{3}{2}(1, -1, 0) + \left(\frac{1}{2}, \frac{1}{2}, -1\right)\right)$.

**Exercise 9.** The edge vectors give $R = R_z(90^\circ)$, so the axis is $(0,0,1)$. But $\mathbf{t} = (0,0,1)$ and $\mathbf{t} \cdot \hat{\mathbf{u}} = 1 \neq 0$. The equation $(I - R)C = \mathbf{t}$ has no solution, since its third row reads $0 = 1$. The motion is a screw motion: a rotation about the $z$-axis combined with a shift of 1 along it.

**Exercise 10.** The top-left entry of the product is $\cos\alpha\cos\beta - \sin\alpha\sin\beta = \cos(\alpha+\beta)$, and the entry below it is $\sin\alpha\cos\beta + \cos\alpha\sin\beta = \sin(\alpha+\beta)$. The rest follows in the same way from the addition formulae. The columns of $R_z(\theta)$ are $(\cos\theta, \sin\theta, 0)$, $(-\sin\theta, \cos\theta, 0)$ and $(0,0,1)$. They have length 1 and are mutually orthogonal, so $R^{\mathsf{T}}R = I$. The determinant is $\cos^2\theta + \sin^2\theta = 1$.

**Summary of formulae**

| | |
|---|---|
| Column rule | $R = [\,R\mathbf{i} \;\; R\mathbf{j} \;\; R\mathbf{k}\,]$ |
| Composition | first $R_1$, then $R_2$: $\;R_2R_1$ |
| Properties | $R^{\mathsf{T}}R = I,\; R^{-1} = R^{\mathsf{T}},\; \det R = 1$ |
| Rodrigues | $R = \cos\theta\, I + \sin\theta\,[\hat{\mathbf{u}}]_\times + (1-\cos\theta)\,\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}}$ |
| Axis through $C$ | $P' = R(P-C)+C,\; M = \begin{bmatrix} R & C - RC \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}$ |
| Angle | $\cos\theta = \frac{1}{2}(\operatorname{tr} R - 1)$ |
| Axis | $2\sin\theta\,\hat{\mathbf{u}} = (R_{32} - R_{23},\, R_{13} - R_{31},\, R_{21} - R_{12})$ |
| Centre | $C_0 = \dfrac{(1-\cos\theta)\mathbf{t} + \sin\theta\,\hat{\mathbf{u}} \times \mathbf{t}}{2(1-\cos\theta)}$, requires $\mathbf{t}\cdot\hat{\mathbf{u}} = 0$ |